



11007 - Characterizing Infrared Stellar Flares on TRAPPIST-1 Using Archival JWST Data

Cycle: 5, Proposal Category: AR

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dra. Guadalupe Tovar Mendoza (PI)	The Johns Hopkins University
Prof. Meredith MacGregor (CoI)	The Johns Hopkins University
Dr. Nestor Espinoza (CoI)	Space Telescope Science Institute
Dr. Rachel A. Osten (CoI)	Space Telescope Science Institute
Dr. Ward Howard (CoI)	University of Colorado at Boulder
Natalie Allen (CoI)	The Johns Hopkins University
Dr. Brett M. Morris (CoI)	Space Telescope Science Institute
Dr. Eric Agol (CoI)	University of Washington
Dr. Hannah Diamond-Lowe (CoI)	Space Telescope Science Institute
Dr. Benjamin Rackham (CoI)	Massachusetts Institute of Technology
Dr. Kevin Stevenson (CoI)	The Johns Hopkins University Applied Physics Laboratory
Dr. Sukrit Ranjan (CoI)	University of Arizona
Dr. Roeland P. van der Marel (CoI)	Space Telescope Science Institute
Mr. Jingcheng Huang (CoI)	Massachusetts Institute of Technology
Dr. Zifan Lin (CoI)	Washington University in St. Louis
Dr. Ana Glidden (CoI)	Massachusetts Institute of Technology

OBSERVATIONS

ABSTRACT

JWST Proposal 11007 (Created: Friday, March 13, 2026, 3:04:33PM Eastern Standard Time) - Overview

Our current understanding of exoplanet atmospheres rests on the assumption that we can disentangle stellar signals from planetary signals in transit transmission data. Recent observations from JWST have shown that stellar contamination dominates the transit transmission data, leaving us with noisy and ambiguous planetary signals. The TRAPPIST-1 system is one of JWST's most studied targets, with over 400 hours of telescope time focused on characterizing the atmospheres of its transiting exoplanets. To improve our understanding of exoplanet systems we will use archival JWST data of TRAPPIST-1 to: (1.) develop a wavelength sensitive flare template to model the morphology of stellar flares in JWST infrared observations and (2.) measure the physical properties of the stellar flares and provide the most complete infrared flare catalog of an M dwarf to date. Both of these products will be made publicly available to the wider scientific community to account for stellar signatures in transmission data for TRAPPIST-1 and other targets.

OBSERVING DESCRIPTION